

Unit - 4

Sustainable Development and Management

Exam-Ready Notes | Environment & Sustainability

1. Climate Change & Ozone Layer Depletion

1.1 Climate Change — Definitions

Climate: Average weather of a particular place over a long period.

Climate Change: Long-term shift in global/regional climate patterns. Caused by human intervention — primarily greenhouse gas emissions and ozone depletion.

Key causes of climate change:

- Presence of greenhouse gases (GHGs) in the atmosphere.
- Depletion of the ozone layer.
- Long-term variation in temperature.

1.2 Ozone Layer — Structure & Function

- Ozone (O₃) is a gas concentrated in the stratosphere, 15–50 km above Earth's surface.
- It absorbs harmful UV (ultraviolet) radiation from the sun, protecting living organisms.
- A 'hole in the ozone layer' occurs when unnatural compounds (free radicals) provide extra catalytic species that enhance ozone destruction.

1.3 Ozone Depleting Substances (ODS)

ODS destroy ozone via catalytic reactions involving free radicals: Cl, Br, H, and NO.

Gas (ODS)	Principal Sources
Chloro Fluoro Carbon (CFC)	Refrigerants, aerosol spray cans
Hydro Chloro Fluoro Carbon (HCFC)	Refrigerants, blowing agents, air conditioners
Bromo Fluoro Carbons (BFC)	Fire extinguishers
Nitric Oxide (NO)	Detonation of nuclear weapons, nitrogenous fertilizers

Remember ODS: 'Can Have Bad Nights'

*C – CFC (aerosol cans, refrigerants)
H – HCFC (air conditioners)
B – BFC (fire extinguishers)
N – Nitric Oxide (fertilizers, nuclear weapons)*

1.4 Impacts of Ozone Layer Depletion

Remember impacts: 'HACE' (Human, Aquatic, Climate, Environment/Materials)

*H – Human health: skin cancer, blindness, reduced immunity
A – Aquatic: kills fauna/flora, affects photosynthesis, causes mutation
C – Climate: global warming, climate change
E – Environmental materials: degradation of paint, plastics*

Effects in Detail:

- 1. Human Health:** UV damages genetic material in skin → skin cancer. Prolonged UV exposure → blindness. Reduced immunity → allergies and infections.
- 2. Aquatic System:** Kills lower fauna and flora. Disrupts photosynthesis and causes mutations.
- 3. Materials:** Degradation of paint quality and plastics.
- 4. Climate:** Overall climate change and global warming.

1.5 Measurement — Dobson Unit (DU)

- Instrument: Dobson Spectrometer.
- 1 DU = 0.01 mm thickness of pure ozone brought to ground level (1 atm).
- Temperate latitudes: ~350 DU | Tropics: ~250 DU.

1.6 Control Measures for Ozone Depletion

1. Reduce CFCs and other ozone-depleting chemicals.
2. Develop and promote alternative chemicals that do not deplete ozone in the stratosphere.

2. Carbon Credit & Carbon Footprint

2.1 Carbon Credit

Carbon Credit: A financial instrument equivalent to one metric tonne of CO₂. Countries/companies that reduce emissions below designated levels can sell 'credits' to high-emitting entities.

- Carbon trading is the central pillar of the Kyoto Protocol — a market-based approach to controlling pollution.

- Carbon trading is emission trading specifically for CO₂, measured in tonnes of CO₂ equivalent (tCO₂e).
- 1 Carbon Credit = 1 Tonne of CO₂.
- High-emission country buys credits; low-emission country sells credits.

Carbon Credit vs Carbon Footprint

Carbon Footprint = TOTAL emissions caused by an activity.

Carbon Credit = REDUCTION in emissions (tradeable).

You buy carbon credits to COMPENSATE for your carbon footprint.

2.2 Carbon Footprint

Carbon Footprint: A measure of the total CO₂ emissions directly and indirectly caused by an activity, or accumulated over the life stages of a product. Usually calculated annually.

Expressed as:

- For products/processes: grams of CO₂ equivalent per kilowatt hour (gCO₂eq/kWh).
- For business: amount of CO₂ emitted directly or indirectly from daily operations.

Fuel Type	CO ₂ Emitted per Litre
Petrol	2.3 kg
Gasoline	2.3 kg
Diesel	2.7 kg
Oil (heating)	3.0 kg

Example: Car consumes 7.5 L diesel per 100 km → 300 km trip = 22.5 L diesel → $22.5 \times 2.7 = 60.75$ kg CO₂.

2.3 Reducing Carbon Footprint (25 Ways — Exam Favourite)

Category Mnemonic: 'TRAVEL FOOD HOME SHOP'

TRAVEL: alternatives to driving, low-carbon vehicles, driving style, tyre inflation, avoid traffic, no excess weight, less air travel, no private jets.

FOOD: local/organic food, avoid imported produce, reduce deforestation.

HOME: insulate home, energy-efficient appliances, LED lighting, smart thermostat, solar panels, water usage.

SHOP: reuse & recycle, support clean energy, fountain pens, reusable bags, print on both sides, vegetable-based cleaners, wash clothes at low temps.

Transport-related (1–8):

- Walk/cycle when possible; use carpooling/public transport.
- Use low-carbon or electric vehicles (no CO2 if charged with clean electricity).
- Smooth driving style — avoid speeding/unnecessary acceleration (reduces mileage up to 33%).
- Inflate tyres properly (+3% mileage); maintain engine, fix oxygen sensors (+40% efficiency).
- Avoid traffic using apps. Remove excess weight; use cruise control.
- Reduce air travel; prefer economy class; use video-conferencing instead.
- Avoid private jets.

Home-related (9–17):

- Insulate and seal home (caulk, weather stripping).
- Use ENERGY STAR-rated appliances.
- LED/CFL bulbs; turn off unused lights.
- Programmable thermostat; avoid extreme settings.
- Install solar panels.
- Reduce water usage (efficient shower heads, drip irrigation, shorter showers).
- Avoid bottled water; use tap water in reusable containers.

Food & Shopping (18–25):

- Eat locally-produced, organic, seasonal food (30% of GHG from food production & transport).
- Reuse and recycle goods (40% of GHG from 'provision of goods').
- Support clean energy (wind, solar, geothermal, hydro, biomass).
- Use fountain pens, not plastic pens.
- Reusable bags for shopping.
- Print on both sides; use recycled inks.
- Vegetable-based (non-oil-derived) cleaning products.
- Wash clothes at low temperatures.

3. Sustainability Practices

3.1 Zero Waste

Zero Waste: A philosophy and design principle for the 21st century. It goes beyond recycling by taking a 'whole system' approach — designing and managing products/processes to reduce volume and toxicity of waste, conserving and recovering all resources, without burning or burying them.

- Aims to imitate sustainable natural cycles where all discarded materials become resources for others.
- Reduces GHGs by: (a) saving energy in extraction/processing/transport, and (b) eliminating landfills and incinerators.

Goals of Zero Waste:

- Maximize recycling
- Minimize waste
- Reduce consumption
- Ensure products are made to be reused, repaired or recycled

- Purchase sustainable products

3.2 Concept of 5R

 **5R Mnemonic: 'Refuse Reduce Reuse Repurpose Recycle' — In Order!**

Before recycling, always go through the first 4 R's.

Think: 'Did I Really Refuse Repurpose?' before putting anything in the recycling bin.

R	Meaning & Action
1. Refuse	Do not buy what you don't really need. Best option — prevents waste at source.
2. Reduce	Reduce garbage generated. Alter lifestyle to minimize waste generation.
3. Reuse	Reuse everything to its maximum after cleaning. Make secondary use of articles.
4. Repurpose	For items that can't be refused/reduced/reused, repurpose them (also called upcycling).
5. Recycle	Recyclable waste to kabadiwallas or composting. Last resort — least preferred.

Note: The 5Rs are ordered by preference — Refuse is the best action, Recycle is the last resort.

3.3 Circular Economy

Circular Economy: A systems solution framework that tackles global challenges like climate change, biodiversity loss, waste and pollution. Products are designed for durability, reuse, and recyclability. Materials for new products come from old products.

- New production and consumption model ensuring sustainable growth over time.
- Optimizes resources, reduces consumption of raw materials, and recovers waste by recycling or repurposing.
- Promotes sustainable development by using waste as an input for producing new finished goods.

Linear Economy vs. Circular Economy:

Linear Economy	Circular Economy
Materials flow: Extract → Make → Dispose	Materials flow: Make → Use → Recover → Remake
Value created by selling maximum products	Value created by keeping materials in use

Leads to resource depletion, excessive pollution	Reduces waste, pollution, ecosystem degradation
Finite — depends on virgin raw materials	Regenerative — uses existing materials
One-directional flow	Closed-loop system

Three Principles of Circular Economy (by design):

- Eliminate waste and pollution
- Circulate products and materials at their highest value
- Regenerate nature

Benefits of Circular Economy:

- Reduces waste by promoting recycling of finished goods.
- Offsets potential price rise of commodities.
- Achieves efficiency and effectiveness through resource recycling.
- Promotes the rental/sharing business model — reuse over purchase.

4. ISO 14001:2004 — Quality System Standard

ISO Purpose: Promote development of standardization and related world activities to facilitate international exchange of goods/services, and to develop cooperation in intellectual, scientific, technological and economic activities.

- The ISO 14000 family addresses various aspects of environmental management.
- ISO 14001:2004 — Provides requirements for an EMS.
- ISO 14004:2004 — Provides general EMS guidelines.

4.1 ISO 14000 Series Standards

Standard	Description
ISO 14001	Environmental management systems — Requirements with guidance for use
ISO 14004	Environmental management systems — General guidelines on principles, systems and support techniques
ISO 14015	Environmental assessment of sites and organizations
ISO 14020–14025	Environmental labels and declarations
ISO 14030	Post-production environmental assessment
ISO 14031	Environmental performance evaluation — Guidelines

ISO 14040–14049	Life Cycle Assessment (LCA) — pre-production planning and environment goal setting
ISO 14050	Terms and definitions
ISO 14062	Making improvements to environmental impact goals
ISO 14063	Environmental communication — Guidelines and examples
ISO 14064	Measuring, quantifying and reducing greenhouse gas emissions



Remember ISO 14000 series — 'Every Assessment Labels Post Evaluations, Life Terms Improve Communication Gas'

14001 – Requirements (EMS)

14004 – Guidelines

14015 – Site Assessment

14020s – Labels

14030 – Post-production

14031 – Performance Evaluation

14040s – Life Cycle Assessment

14050 – Terms

14062 – Improve impact goals

14063 – Communication

14064 – GHG quantification

4.2 Uses of ISO 14001:2004 (EMS)

An EMS meeting ISO 14001:2004 enables an organization to:

- Identify and control the environmental impact of its activities, products or services.
- Improve its environmental performance continually.
- Implement a systematic approach to setting environmental objectives and targets, and demonstrating they have been achieved.

5. Environment Impact Assessment (EIA)

EIA: A formal system to assess the environmental consequences of developmental activities before any project is undertaken, to reduce or eliminate adverse effects on biological, chemical, physical, social and economic aspects of the environment.

- EIA report is often a key requirement for gaining Environmental Clearance (EC) for development projects.
- Formally recognized at the Rio Earth Summit (1992). Principle 17 of Rio Declaration: 'EIA shall be undertaken for proposed activities likely to have significant adverse impact on the environment.'

5.1 Objectives of EIA

- Identify potential impacts of a proposed project at an early stage.
- Assess the possibility of impact mitigation.
- Recommend preventive and mitigating measures to authorities.
- Provide unbiased information to decision-makers about environmental implications.

5.2 Steps in EIA

 **Remember 11 steps: 'Scoping Collects Assessments, Evaluations Analyse Delineations, Specifications, Environmental Documents, Reviews Decisions'**

S – Scoping

C – Collection of Baseline Data

A – Assessment of Environmental Impact

E – Evaluation of Impact Assessment

A – Analysis of Alternatives

D – Delineation of Mitigation Procedures

S – Specification of Environmental Monitoring Programme

E – Environmental Management Plan

D – Documentation

R – Review of Report

D – Decision-making

Step	Description
1. Scoping	First step — identify all important issues; hold discussions with stakeholders; select primary impacts based on magnitude and eco-sensitivity.
2. Baseline Data Collection	Collect physical, biological, socio-economic data within and around project site. In India: 10–25 km radius depending on sensitivity.
3. Assessment of Impact	Identify, predict and assess positive/negative impacts (air quality, noise, wildlife, social systems, employment).
4. Evaluation	Determine if predicted impacts are significant enough to require alternatives or mitigation. Consult laws, regulations, eco-sensitivity.
5. Analysis of Alternatives	Compare feasible alternatives (site, technology, design, operation) against 'without project' scenarios.
6. Mitigation Procedures	Detail measures: pollution controls, waste treatment, compensation, restoration, off-site programmes.
7. Monitoring Programme	Specify monitoring measures during construction and operation to reduce adverse impacts.
8. Environmental Management Plan	Elaborate management procedures using cost-benefit analysis and pair-wise comparisons.

9. Documentation	Draft EIA report: project description, executive summary, resource issues, data gaps, consultant disclosure, public summary.
10. Review of Report	Evaluate quality and efficiency of EIA; incorporate expert suggestions.
11. Decision-making	Decision-makers determine project prospects based on EIA review recommendations.

5.3 Benefits & Limitations of EIA

Benefits	Limitations
Reduced project implementation time	Limited range of alternatives considered
Cost-saving modifications possible at start	Lack of baseline environmental database
Sustainable resource utilization	Complicated — requires multi-discipline experts
Fewer conflicts of interest	Improved methodologies needed for better outcomes
Compliance with laws and regulations	Reports are too academic, technical and lengthy
Increased public project acceptance	EIA often ends after project clearance — no follow-up

6. Environment Management Plans (EMP)

EMP: A formal system with a set of management procedures allowing an organization to identify, evaluate and reduce the environmental impact of its activities, products and services — in a planned, formal, comprehensive, systematic and documented manner.

6.1 Objectives of EMP

- Restrict and regulate overexploitation of natural resources.
- Renew natural resources with suitable devices.
- Set targets to reduce energy, water use, and landfill waste.
- Set environment-friendly purchasing procedures.
- Improve management of environmental impacts.
- Define key responsibilities for achieving targets.
- Monitor and measure environmental performance with key indicators.
- Ensure compliance with legal and environmental requirements.
- Build goodwill among customers, employees and stakeholders.

- Shift from traditional management culture to environmental-organization culture.

6.2 Implementation of EMP — 5 Components

Component	Key Points
1. Environmental Policy	Written document; appropriate to organization's activities; commitment to continuous improvement; compliant with laws; lays down framework of environmental objectives.
2. Planning	Identifies environmental aspects, legal requirements, objectives/targets; develops environmental management programme.
3. Implementation & Operation	Mobilizes resources; covers HR management, training, documentation, communication, operational control, emergency preparedness.
4. Monitoring & Evaluation	Identifies performance indicators; ensures compliance; periodic audits by trained internal/external auditors.
5. Management Review	In-depth analysis going beyond compliance; addresses audit recommendations, new regulations, stakeholder interests, public pressures.

Note: ISO 14000 provides internationally recognized standards for structuring the EMP.

7. Green Business & Eco-Labeling

7.1 Green Business

Green Business: A sustainable business that has minimal negative impact on the global or local environment, community, society or economy. Adopts principles, policies and practices that improve the quality of life for customers, employees, communities and the planet.

4 Criteria for a Green Business:

- Incorporates principles of sustainability into each business decision.
- Supplies environmentally friendly products/services that replace non-green demand.
- Is greener than traditional competition.
- Has made an enduring commitment to environmental principles in its operations.

7.2 Eco-Labeling

Eco-Label: A voluntary method of environmental performance certification and labelling. Identifies products/services proven environmentally preferable overall, within a specific category.

Three ISO Types of Eco-Labels:

Type	Description
Type I	Voluntary, multiple-criteria based, third-party programme that awards a license for use of environmental labels indicating overall environmental preferability.
Type II	Informative environmental self-declaration claims.
Type III	Voluntary programmes providing quantified environmental data under pre-set categories set by a qualified third party.

Benefits of Eco-Labeling:

- Informs consumer choice about environmental impacts.
- Promotes economic efficiency — cheaper than regulatory controls.
- Stimulates market development toward greater environmental awareness.
- Encourages continuous improvement in environmental performance.
- Promotes certification — a seal of approval.
- Assists in monitoring environmental claims.

Ecomark (India): Launched in 1991 by the Government of India. Uses a cradle-to-grave approach (raw material extraction → manufacturing → disposal). Awarded to consumer goods meeting specified environmental criteria and Indian Standards quality requirements.

8. Green Buildings & Sustainable Habitat

8.1 Sustainable Habitat

Sustainable Habitat: An approach towards balanced and sustainable development of the ecosystem offering adequate shelter with basic services, infrastructure, livelihood opportunities, environmental and socio-economic safety including equality, inclusiveness and disaster-resilience.

8.2 Green Building

Green Building: An approach to creating a built environment on principles of sustainability — uses local/recycled materials, less water, less grid energy dependence, less waste generation, and provides healthy living conditions in harmony with the environment.

- Designed to reduce overall impact on human health and the natural environment.
- Requires careful, systemic attention to full life-cycle impacts of resources embodied in the building.

8.3 Components (Principles) of Green Building

Component	Detail
1. Energy Demand	Minimize grid dependence via natural ventilation, solar passive heating, passive lighting, energy-saving roofs.
2. Water Demand	Reduce water demand using water-efficient products and services.
3. Low Energy Building Materials	Use reused/recycled, locally available, nontoxic, recyclable materials.
4. Waste Reduction	Reduce waste from construction through renovation and demolition.
5. Indoor Air Quality	Provide proper ventilation; prevent dampness to avoid fungi/algae.
6. Harmony with Environment	Minimum alteration to environment; blend with aesthetics of surroundings.

8.4 Benefits of Green Buildings

Environmental Benefits:

- Reduced indoor pollution.
- Reduced water consumption → water conservation.
- Reduced energy consumption → lower carbon footprint.
- Low environmental alteration.
- Limited waste generation during construction, occupation and demolition.

Economic Benefits:

- Cost savings from reduced water and grid energy use.
- Enhanced image and marketability.
- Cost savings from low-energy building materials.

8.5 Green Building Certification Systems

System	Origin
GRIHA — Green Rating for Integrated Habitat Assessment	India
CASBEE — Comprehensive Assessment System for Building Environmental Efficiency	Japan
LEED — Leadership in Energy and Environmental Design	USA



Remember certification systems: 'GCL — Go Clean Living'

G – GRIHA (India)

C – CASBEE (Japan)

L – LEED (USA)

8.6 Planning for Sustainable Building

- Green building materials: bamboo, straw, recycled stone, recycled metal, sheep wool, compressed earth blocks, cork, etc.
- Reduced energy use: orient windows/walls to maximize solar gain in winter and minimize in summer.
- Reduced waste: minimize material going to landfills during construction.
- Rainwater harvesting: collect from terrace/roof and store in underground tanks.
- Reduction of pollution: human waste to semi-centralized biogas plant → produces liquid fertilizer.

9. Energy Efficiency

Energy Efficiency: The use of energy in an optimum manner to achieve the same service that could have been achieved using a common less efficient manner. Practice of reducing energy requirements while achieving the required energy output.

- Energy-efficient homes use less energy to heat, cool and run appliances.
- Energy-efficient manufacturing uses less energy to produce goods.
- One of the easiest and most cost-effective ways to combat climate change.
- Vital component in achieving net-zero emissions through decarbonization.

Why is it important? (2M answer):

- Reduces energy costs for consumers.
- Reduces greenhouse gas emissions and combats climate change.
- Conserves finite natural resources.
- Aids in achieving net-zero carbon emissions (decarbonization).

10. Sustainable Transport

Sustainable Transport: Any means of transportation that is green, has low impact on the environment, and does not rely on the world's finite natural resources. Also known as Green Transport. Runs on renewable/regenerated energy rather than fossil fuels.

- Examples: walking, cycling, sailing, transit, carpooling, car sharing, and green vehicles.
- Sustainable transportation options run on clean fuel, batteries or both.

Alternative Fuels:

Biodiesel, Electricity, Ethanol, Hydrogen, Natural Gas, Propane.

 **Remember alternative fuels: 'BEEH NP'**

B – Biodiesel
E – Electricity
E – Ethanol
H – Hydrogen
N – Natural Gas
P – Propane

Benefits of Sustainable Transport:

- Reduced traffic congestion.
- Cost savings on fuel and vehicles.
- Reduced greenhouse gas emissions.
- Reduced dependence on non-renewable energy.
- Reduced transportation costs.
- Increased physical activity.
- Increased social interaction.
- Support for local businesses and vibrant economy.
- Healthier lifestyles and better quality of life.
- Improved accessibility to reliable, affordable transport for all.

11. Sustainable Energy — Non-Conventional Sources

Non-Conventional Energy: Sources of energy produced continuously in nature that are inexhaustible. Also known as renewable energy. These replace conventional (non-renewable) sources whose known reserves have been depleted.

Forms of Renewable Energy:

 **Remember: 'SWBHW — Solar Waves Blow Hydro Geo Wind'**

S – Solar energy
W – Wind energy
B – Bio energy
H – Hydro energy
G – Geothermal energy
W – Wave and tidal energy

12. Carbon Sequestration

Carbon Sequestration: The process of capturing, securing and storing carbon dioxide from the atmosphere with the goal of reducing global climate change.

- ~45% of CO₂ emitted by humans remains in the atmosphere — a major driver of global warming.
- Carbon Capture and Storage (CCS): capture CO₂ from industrial processes (coal plants) and store it in deep geological formations.
- Aims to stabilize carbon in solid and dissolved forms so it doesn't warm the atmosphere.

Two Main Types:

1. Biological Carbon Sequestration (Indirect/Passive)

Carbon stored in natural environments — carbon sinks:

Carbon Sink	How it Works
Forests	CO ₂ binds to plants during photosynthesis. Forests store 2x more carbon than they emit. 25% of global emissions sequestered in forests/grasslands.
Soil	Through bogs, peat and swamps — carbon stored as carbonates with Ca/Mg over thousands of years. Carbon held for 70,000+ years in some cases.
Oceans	Absorb another ~25% of emitted CO ₂ . Mostly in upper ocean layers. Excess CO ₂ can acidify water, threatening biodiversity.

2. Geological Carbon Sequestration (Artificial/Direct)

- Carbon stored in underground geological formations or rocks.
- Graphene production: requires CO₂ as raw material.
- Engineered molecules: scientists modify molecules to form new compounds by capturing atmospheric carbon.
- Carbon Capture and Storage (CCS): CO₂ from power generation/industry is compressed and injected into deep rock formations for permanent storage.

13. Green Engineering

Green Engineering: The design, commercialization and use of processes and products to reduce pollution, promote sustainability and minimize risk to human health and the environment, without sacrificing economic viability and efficiency.

- Combines environmentally conscious attitudes, values and principles with science, technology and engineering practice.

- Greatest impact when applied early, in the design and development phase.
- Necessarily interdisciplinary — applicable across all engineering disciplines.

13.1 Processes in Green Engineering

Process	Description
Waste Reduction	Minimize waste from manufacturing/shipping; find new fuel methods; reduce unnecessary production steps.
Materials Management	Find better, safer materials for product design/manufacturing; invent more efficient production methods.
Pollution Prevention	Identify pollution sources; integrate cleaner techniques; enhance manufacturing and delivery cleanliness.
Product Enhancement	Improve products while making them safer for the environment; find alternate energy sources and greener manufacturing.

13.2 Twelve Principles of Green Engineering



Remember first 6: 'I P D M O C' — 'I Prevent Design, Maximize Output, Conserve'

1. *Inherent Rather Than Circumstantial*
2. *Prevention Instead of Treatment*
3. *Design for Separation*
4. *Maximize Efficiency*
5. *Output-Pulled vs Input-Pushed*
6. *Conserve Complexity*

Principle	Statement
1. Inherent Rather Than Circumstantial	All materials and energy I/O should be as inherently non-hazardous as possible.
2. Prevention Instead of Treatment	Better to prevent waste than to treat it after it is formed.
3. Design for Separation	Separation/purification operations should minimize energy and materials use.
4. Maximize Efficiency	Products/processes/systems should maximize mass, energy, space and time efficiency.
5. Output-Pulled vs Input-Pushed	Systems should be output-pulled, not input-pushed through energy and materials.
6. Conserve Complexity	Embedded entropy and complexity should be considered as an investment in design choices.

7. Durability Rather Than Immortality	Targeted durability — not immortality — should be the design goal.
8. Meet Need, Minimize Excess	'One size fits all' solutions should be considered a design flaw.
9. Minimize Material Diversity	Reduce material diversity in multicomponent products to promote disassembly and value retention.
10. Integrate Material and Energy Flows	Product/process design must include integration with available energy and material flows.
11. Design for Commercial Afterlife	Products/processes/systems should be designed for performance in a commercial afterlife.
12. Renewable Rather Than Depleting	Material and energy inputs should be renewable rather than depleting.

13.3 Cradle to Cradle (C2C) Concept

Cradle to Cradle (C2C): A way of designing products or processes that work like natural systems — where every material is recycled into a new product at the end of its life, resulting in zero waste. Replaces the make-take-dispose approach.

- C2C design: products developed for closed-loop systems where every output is safe — either biodegrading naturally (biological nutrient) or being fully recycled into high-quality materials (technical nutrient).
- Aims to minimize environmental impact through sustainable production, operation, and disposal.
- Incorporates social responsibility into product development.

14. Urbanization & Low Carbon Life Cycle

14.1 Urbanization

Urbanization: A process of moving population from rural areas to urban areas for improving life standards and lifestyle through scientific and technological developments.

Energy Problems Due to Urbanization:

- Pollution from coal.
- Acid rain — SO₂, NO_x from industries react with water to form sulphuric acid.
- Vehicle pollution — exhaust from two-wheelers, four-wheelers, transport.
- Deforestation — for housing and timber.
- Global warming — fossil fuel combustion creates greenhouse effect.
- Increased electricity demand — AC, washing machines, refrigerators, water heaters.

Solutions for Urban Energy Problems:

- Use public transport instead of private vehicles.
- Reduce energy consumption in all respects.
- Use energy-efficient devices.
- Encourage use of solar and wind energy.
- Impose strict laws, penalties and energy audits.

14.2 Low Carbon Life Cycle

Low Carbon Life: A lifestyle in which people minimize energy consumption and carbon greenhouse gas emissions. Low carbon economy = model based on low energy consumption, low pollution, and low emission.

- Activities creating CO₂: driving, heating homes, generating electricity, flying, manufacturing goods, long-distance transport.
- Low carbon lifestyle directly counters these through sustainable choices.

Quick Reference — Key Definitions (for 10M define questions)

Term	One-Line Definition
EIA	Formal system to assess environmental consequences of developmental activities before any project is undertaken.
Climate Change	Long-term shift in Earth's climate patterns due to human intervention — primarily GHG emissions and ozone depletion.
Carbon Credit	A financial instrument equal to 1 metric tonne of CO ₂ — tradeable right to emit (Kyoto Protocol mechanism).
Carbon Footprint	Total CO ₂ emissions directly and indirectly caused by an activity or accumulated over a product's life stages.
Green Building	Built environment on sustainability principles — uses recycled materials, less water/energy, minimal waste, healthy living conditions.
Zero Waste	Philosophy to design/manage products so that all resources are conserved and recovered without burning or burying.
Circular Economy	Economic model where products are designed for durability, reuse and recyclability — materials from old products go into new ones.
Carbon Sequestration	Process of capturing and storing CO ₂ from the atmosphere to reduce climate change.
Energy Efficiency	Use of energy in optimum manner to achieve the same service using less energy.

Sustainable Transport	Any transportation that does not rely on finite natural resources — uses renewable/regenerated energy.
Eco-Labeling	Voluntary certification method identifying products/services with proven environmental preferability.
Green Engineering	Design and use of products/processes to reduce pollution and minimize risk to health and environment without sacrificing efficiency.
Dobson Unit (DU)	Unit of measurement for atmospheric ozone — 1 DU = 0.01 mm pure ozone thickness at ground level pressure.
Ecomark	India's eco-labelling scheme launched in 1991 using cradle-to-grave approach for consumer goods.
EMP	Environment Management Plan — formal system to identify, evaluate and reduce environmental impact of an organization's activities.